

is also setting a new benchmark for inclusivity in software development. Their tools are thoughtfully designed to empower users by focusing on accessibility, adaptability, and ease of use. Let's explore how Zendalona's mission aligns with the FOSS ethos and delve into their transformative suite of products.

## Zendalona: Revolutionising accessibility through software

Zendalona develops free and open source tools tailored to meet the unique needs of visually impaired users. By leveraging FOSS principles, Zendalona ensures transparency, adaptability, and community-driven enhancements in their offerings.

### Accessible-Coconut (AC)

**Description:** Accessible Coconut is a GNU/Linux-based operating system designed to cater to the needs of visually impaired users by integrating accessibility features out-of-the-box. It comes with a pre-installed screen reader software for immediate usability, customisable keyboard shortcuts tailored for accessibility, and is compatible with Braille devices and speech output systems.

**Benefits:** Accessible Coconut provides a seamless computing experience for visually impaired users, eliminating the need for additional software installations and ensuring a user-friendly interface right from the start. While making technology accessible to all, it fosters digital inclusion.

### Sharada-Braille-Writer (S.B.W.)

**Description:** Sharada Braille Writer is a Linux-based text editor that allows Braille input using Perkins-like methods. It

simplifies multi-language text creation for visually impaired users. It supports Braille input for over 40 languages including several Indian languages such as Hindi, Malayalam, Kannada, Arabic, etc; utilises key combinations corresponding to Braille dots for intuitive typing, and is compatible with screen readers for seamless interaction.

**Benefits:** S.B.W. empowers visually impaired individuals to create, edit, and share text in their native languages efficiently, fostering independence and digital literacy.

### Linux-Intelligent-OCR-Solution (LIOS)

**Description:** Lios converts printed or digital text into accessible formats using OCR technology, designed with accessibility at its core. It supports text extraction from scanned images, PDFs, and screenshots, has integrated camera and scanner functionality, and is fully compatible with screen readers.

**Benefits:** Lios bridges the gap between printed materials and accessible formats, enabling visually impaired users to engage with a broad range of texts for education and daily tasks.

### IBus-Braille

**Description:** IBus-Braille enables Braille typing on graphical desktops using Perkins-style keyboard input, with support for numerous Braille tables. It contains customisable key mappings for user-specific preferences, integration with Liblouis for accurate Braille translations, and one-hand typing support with Three to Six mode.

**Benefits:** This tool simplifies Braille-based typing for users, enhancing productivity and reducing the learning curve for interacting with digital environments.

### Accessible games: TuxMath and TuxType Extensions

**Description:** These educational games are adapted for visually impaired users, combining entertainment with skill development. They have audio feedback and Braille support for interactive gameplay, math drills and typing practice tailored for accessibility, and multiplayer and single-player modes.

**Benefits:** These games make learning enjoyable and inclusive, helping users develop essential skills while having fun.

Zendalona stands at the forefront of accessibility innovation, blending FOSS principles with a commitment to inclusivity. Their software does more than address the needs of the visually impaired; it also creates opportunities for empowerment and growth in a digital-first world. By fostering collaboration and adaptability, Zendalona transforms barriers into bridges, equipping users with tools that enable education, employment, and personal growth.

Through their community-driven approach, Zendalona exemplifies how technology can be a force for good, championing equity and ensuring no one is left behind in the digital age. By combining innovation with compassion, Zendalona not only redefines accessibility but also paves the way for a future where inclusivity is the standard, not the exception. 

## Reference

- [Zendalona's projects at zendalona.com](https://zendalona.com)

## By: Nigyaasha Shrivastava

The author is a tech enthusiast and a volunteer with Zendalona. She currently works as a value creation advisor for private equity in a global technology consulting firm.